

Lab 3: Architecture Selection & Justification

Dropshipping Inventory & Order Orchestrator

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Architecture Selection

We chose Microservices Architecture for the Dropshipping Inventory & Order Orchestrator.

Reason 1

The system has several separate responsibilities, including receiving storefront webhooks, verifying supplier inventory, splitting multi-vendor orders, dispatching supplier purchase orders, and tracking shipments. Microservices allow these responsibilities to be separated into focused services, making the system easier to maintain and modify.

Reason 2

The system depends on external storefront and supplier APIs, which may have different requirements or temporary failures. With microservices, individual services can handle these dependencies independently, reducing the impact of a problem in one part of the system on the rest of the order-processing workflow.

Security Advantage

Microservices allow sensitive functionality to be isolated. The webhook service can authenticate and validate incoming requests, while payment-related operations can be handled separately. This limits unnecessary access between components and helps protect sensitive operations.

Performance Benefit

Individual services can be scaled according to their workload. For example, the webhook and order-processing services can handle increased order traffic without requiring every service to scale equally. This supports faster processing and helps meet the requirement for quick webhook acknowledgement.